

MISSION PLANNER + SITL

Beginner's guide to the interface, UAV terminology, mission concepts, and simulated flight on Windows

A domain-knowledge guide for exploring ArduPilot SITL through Mission Planner. It explains what the controls and telemetry mean; it is not a real-flight checklist.

Important: This guide describes the common Copter workflow. Mission Planner menus and available options can vary by firmware and version. Use simulation only for learning; do not treat simulated behavior as proof that a real aircraft is safe.

1. What are Mission Planner, ArduPilot, and SITL?

Term	Meaning
Mission Planner	A Windows ground control station (GCS): it displays vehicle data, maps, missions, parameters, and setup tools.
ArduPilot	Autopilot firmware family. ArduCopter is for multirotors, ArduPlane for fixed-wing aircraft, and ArduRover for ground vehicles.
SITL	Software In The Loop: the autopilot runs as a software process on the computer and simulates a vehicle, instead of controlling physical flight hardware.
GCS	Ground Control Station: software used to monitor and interact with a vehicle.
MAVLink	A messaging protocol commonly used between autopilots, simulators, and ground stations.
Telemetry	Vehicle status and sensor/navigation data sent to the GCS.
Firmware	The autopilot software for a vehicle type; it determines supported modes, commands, and parameters.

2. Typical Windows SITL learning setup

The simple learning flow is: install or obtain Mission Planner → start a compatible ArduPilot SITL simulator → connect Mission Planner to the simulator's MAVLink endpoint → inspect telemetry and map → explore flight modes → optionally create a small simulated mission.

Connection concept: SITL acts as the virtual vehicle; Mission Planner is the operator interface. They exchange MAVLink messages over a local network connection, commonly UDP. The exact port and launch method depend on the simulator package and instructions.

Component	Role
Windows PC	Runs the simulator and Mission Planner.

Component	Role
SITL process	Runs simulated autopilot logic and virtual vehicle dynamics.
MAVLink connection	Carries telemetry and commands between SITL and Mission Planner.
Mission Planner	Shows the HUD/map, lets you inspect modes and mission items, and displays messages.

Connection troubleshooting: confirm the simulator is running; check the connection type and port shown by the SITL instructions; use the matching Mission Planner connection option; allow the application through Windows Firewall only when you understand the network scope; check that the simulator and GCS are configured for compatible MAVLink endpoints.

3. Main Mission Planner screens

Flight Data

The live monitoring workspace. It commonly contains a map, HUD/instrument display, status/messages, and vehicle controls.

UI term	Meaning
HUD	Heads-Up Display: instrument-style view of vehicle attitude, heading, speed, and altitude.
Artificial horizon	Visual representation of pitch and roll.
Pitch / Roll / Yaw	Rotation about the vehicle's lateral / longitudinal / vertical axes (convention depends on frame).
Heading	Direction the nose/front of the vehicle points, usually shown in degrees.
Groundspeed	Speed relative to the ground.
Climb rate / VSpeed	Vertical rate: positive generally means climbing; negative means descending.
Map / track	Vehicle location and recorded or current path.
Messages / Messages tab	Text status, warnings, notices, and events from the autopilot.
Flight mode	Current autopilot control mode, such as STABILIZE, LOITER, or AUTO.

Flight Plan

The mission-planning workspace. You can view a map, add mission items, set coordinates and altitude, and read or write mission data when connected.

Term	Meaning
Mission item / command	One step in a mission: a navigation point or an action.

Term	Meaning
Waypoint (WP)	A navigation target the vehicle is instructed to reach.
TAKEOFF	Mission command to begin an automated takeoff, where supported.
LAND	Mission command to perform a landing sequence.
RTL	Return to Launch: return toward the configured home/return point and usually land, depending on settings.
Loiter	Remain near a location or hold position for a specified condition/time.
Read / Write	Read mission data from the vehicle / upload mission data to it.
Altitude field	Altitude value for an item; its reference depends on command and configuration.

Initial Setup and Config/Tuning

Setup areas contain vehicle setup and calibration workflows. Config/Tuning exposes parameters and other behavior settings. In SITL, these screens are useful for learning terminology; changing parameters can alter simulated behavior.

Term	Meaning
Compass	Magnetic-field sensor used for heading estimation; simulation may model sensor data.
Accelerometer	Measures specific force and helps estimate attitude and motion.
Gyroscope	Measures angular rate (rotation rate).
Calibration	Procedure to estimate sensor offsets/scales; exact relevance differs in simulation.
Parameter	Named setting that controls some aspect of firmware behavior.
EKF	Extended Kalman Filter: estimator that combines sensor data to estimate attitude, position, and velocity.
Arming	State in which the autopilot permits motor/actuator output; SITL arming is simulated.
Pre-arm checks	Conditions checked before arming is allowed.

4. Flight modes (Copter examples)

Mode	Plain-language concept
STABILIZE	Pilot input controls attitude; autopilot helps stabilize the vehicle. It does not automatically hold position.

Mode	Plain-language concept
ALT HOLD	Helps maintain altitude while the pilot controls horizontal movement.
LOITER	Uses navigation data to hold position and altitude when conditions and sensors permit.
AUTO	Runs the uploaded mission automatically.
GUIDED	Accepts guided commands such as moving to a target location.
RTL	Returns to the configured return/home location using configured behavior.
LAND	Runs the landing behavior.
POSHOLD	Position-holding mode with pilot control characteristics that differ from LOITER.

Mode names and behavior differ across ArduPilot vehicle types and firmware versions. Do not assume a Copter mode exists or behaves identically on Plane or Rover.

5. Altitude, position, and navigation vocabulary

Term	Meaning
AGL	Above Ground Level: height above the local ground directly below or near the vehicle.
AMSL	Above Mean Sea Level: altitude referenced to mean sea level.
ASL	Above Sea Level: commonly used informally for altitude above sea level; AMSL is the more precise term.
Relative altitude	Altitude relative to a reference point, often home/takeoff location in a GCS display.
Home	Reference location recorded for return/navigation functions; it is not necessarily identical to the current position.
Latitude / Longitude	Geographic coordinates, typically expressed in degrees.
GPS fix	Status of satellite-based positioning; a fix indicates the receiver has a position solution.
HDOP	Horizontal Dilution of Precision: an indicator of satellite geometry affecting horizontal positioning precision; lower is generally better.
Ground track	Direction of movement over the ground; it can differ from heading when wind or sideways motion is present.

Example: If the home point is 200 m AMSL and a vehicle is 50 m above that point, its altitude is approximately 250 m AMSL, assuming the reference and terrain are consistent. AGL is measured from local terrain and may differ.

6. Mission concepts

Concept	Meaning
Mission sequence	Ordered list of commands the autopilot executes in AUTO mode.
Takeoff item	Starts the mission with a takeoff action where supported.
Navigation item	Moves the vehicle toward a location or route point.
Action item	Performs an operation such as changing speed, waiting, or landing.
Waypoint radius / acceptance	How close the vehicle must get for a waypoint to count as reached; exact behavior depends on configuration.
Loiter time	Duration associated with a loiter command.
Speed command	Mission instruction that changes target speed, when supported.
Failsafe	Configured response to a condition such as lost RC, low battery, or lost communication; behavior depends on settings.
Geofence	Virtual boundary used to constrain or monitor vehicle movement.

7. Telemetry, status, and common messages

Term	Meaning
Heartbeat	Periodic MAVLink message indicating a component is present and communicating.
Link / connection status	Whether Mission Planner is receiving data from the simulated vehicle.
GPS: 3D Fix	GPS receiver reports a 3D position solution; it does not by itself guarantee navigation quality.
EKF variance / EKF status	Estimator consistency/health indicators; interpret in context of the message and firmware.
Battery voltage	Electrical potential reported by the simulated battery model.
Current / consumed mAh	Estimated current draw / accumulated battery capacity used, if modeled.
Vibration	Reported vibration metrics; simulator values may be idealized.
Failsafe triggered	A configured safety response has activated; read the full message to learn which condition.

Term	Meaning
PreArm: ...	A reason arming is blocked until a check is satisfied or bypassed by supported configuration.

8. A simple SITL exploration session

1. Start the ArduPilot SITL simulator using a supported Windows setup and choose a vehicle model such as Copter.
2. Start Mission Planner and connect using the connection type/endpoint specified by that simulator setup.
3. In Flight Data, locate the HUD, map, mode display, GPS status, altitude, speed, and message area.
4. Observe the simulated vehicle's initial position and home reference. Distinguish heading from ground track and AGL from AMSL/relative altitude.
5. Explore available flight modes in simulation and read the mode descriptions. Observe how the HUD and map change.
6. Open Flight Plan and inspect the structure of mission items: takeoff, waypoints, loiter/action, and landing/return commands.
7. If you choose to run a mission, use a short, uncomplicated simulated mission and follow the simulator's own instructions for uploading and starting it.
8. Review messages and telemetry during the simulation. Stop the simulator when finished.

Learning-only caution: Do not use this guide as instructions to arm or operate a physical aircraft. Real vehicles require appropriate training, legal compliance, site safety, hardware checks, and a separately validated procedure.

9. Quick glossary

Abbreviation	Expansion / meaning
GCS	Ground Control Station
SITL	Software In The Loop
HITL	Hardware In The Loop
UAV / UAS	Unmanned Aerial Vehicle / Unmanned Aircraft System
GCS	Ground Control Station
GPS / GNSS	Global Positioning System / Global Navigation Satellite System
IMU	Inertial Measurement Unit; typically accelerometers and gyroscopes, sometimes additional sensors
EKF	Extended Kalman Filter
AGL / AMSL	Above Ground Level / Above Mean Sea Level
RTL	Return to Launch
WP	Waypoint
MAVLink	Micro Air Vehicle Link messaging protocol

10. Official references

Use the official documentation for current installation steps, connection details, and version-specific behavior:

ArduPilot Mission Planner: <https://ardupilot.org/planner/>

Mission Planner Flight Plan: <https://ardupilot.org/planner/docs/mission-planner-flight-plan.html>

Mission Planner features: <https://ardupilot.org/planner/docs/mission-planner-features.html>

ArduPilot SITL overview: <https://ardupilot.org/dev/docs/sitl-simulator-software-in-the-loop.html>

Mission Planner simulation guide: <https://ardupilot.org/planner/docs/mission-planner-simulation.html>

This guide is a beginner-oriented terminology reference. Menus, available modes, and simulator connection settings can change by software version and vehicle firmware.